Mohammed Abdul Aleem Hazari

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.54
Sri Sanjeevni Junior College	Class XII (Telangana State Board of Intermediate Education)	2020	97.00%
International Indian Public School, Riyadh	Class X (Central Board of Secondary Education)	2018	90.40%

ACADEMIC ACHIEVEMENTS

- **Academic Excellence Award in IITH 2026**
- Secured **First Place** at ELEC TRIVIA Technical Competition, NIT Puducherry
- Secured **Rank 2** in **National Science Olympiad** in the School level conducted by Science Olympiad Foundation

POSITIONS OF RESPONSIBILITY

- **Placement Coordinator, IITH**
- **Class Representative, IITH & EEE NIT-PY**
- Event Coordinator at NITPY Cultural Fest & Technical Fest
- Joint Secretary, SPREE (EEE ASSCOCIATION), NITPY

SKILLS

- **Programming & Tools:** Verilog HDL, C/C++, AMD Vivado, Cadence Virtuoso, SimVision, COMSOL Multiphysics, ANSYS (Electronics, Icepack, SIwave), Altium Designer, LTspice, MATLAB, PCB design, Git, GitHub, Linux(basic)

RELEVANT COURSES

IC design: Digital IC Design, CMOS Digital Design, VLSI Technology, Computer organization architecture , Analog IC Design

IC Packaging: IC Assembly, Packaging Design and Characterization, Heat Transfer in Electronic Packaging, Fatigue and Fracture in Electronic Packaging, Introduction to Advanced Packaging

PROJECTS

M.tech thesis *(Ongoing)* -AI/ML-Driven Thermal-Aware Co-Optimization Framework for TSV-Based 3D Packaging | IIT Hyderabad x Google Research | *COMSOL Multiphysics*

- Simulated heat transfer in 3D stacked **Qubit-Chip/Interposer structures** (QUIP) using **COMSOL Multiphysics** to mitigate thermal noise in superconducting quantum annealing machines.
- Executed large-scale **parametric sweeps** adjusting TSV side lengths and pitches, identifying geometric bottlenecks

- Bypassed expensive computational FEM runs with a <5% prediction error margin by validating a **multi-hotspot** thermal superposition theorem to train predictive AI/ML models

Design of a Processor based on Harvard Architecture | *Verilog HDL, AMD Vivado*

- Designed a 16-bit Harvard processor in Verilog utilizing a **6-state FSM control unit**, implementing a **32-bit ISA** supporting **25+ operations** across both **3-address register** and **immediate data addressing modes.**
- Enabled complex hardware-level looping, dynamic **jump instructions** triggered by hardware-level **ALU status flags.**
- Verified 100% cycle-accurate functionality and **external I/O data routing** by simulating custom assembly **testbenches** within **Xilinx Vivado** to confirm overall hardware integrity and seamless internal **data path synchronization.**

Design of Parameterized Asynchronous CDC FIFO | *Verilog HDL, AMD Vivado*

- Designed a 100% scalable **Clock Domain Crossing FIFO buffer** preventing data loss across asynchronous systems by developing parameterized **Verilog RTL** tailored for high-speed digital ASIC design.
- Resolved **metastability** issues ensuring zero data corruption during **asynchronous transfers** by deploying standard N+1 bit **Gray code conversion** alongside robust **dual-stage flip-flop synchronizers.**
- Optimized system performance to achieve **1-cycle flag latency** preventing memory overflows by **look-ahead pointer** combinational logic for **empty and full flag generation.**

Design and analysis 6T SRAM using 180nm | *Cadence Virtuoso*

- Designed **schematic** and an area-optimized **layout** for a **6T SRAM cell** in Cadence Virtuoso using **180nm CMOS technology node**.
- Performed **DC analysis** to extract and analyze **butterfly curve** for evaluating cell stability and Static Noise Margin (SNM).
- Verified design integrity by successfully running **Design Rule Checks (DRC)** and **Layout Versus Schematic (LVS**) checks to ensure physical design matched the electrical schematic without violations.

High-Frequency Low Noise Amplifier (LNA125) Design | *Altium Designer* | *Ansys SIwave* | *PCB Design*

- Designed and tested a 40x30 mm HF LNA (PGA-103+) in **Altium Designer,** integrating discrete components to achieve a **~20dB peak gain** across the **100-150 MHz** target band.
- Tailored RF trace widths to PCB Power's layer stackup to achieve **50-ohm** impedance alongside extensive **ground via-stitching.**
- Validated signal integrity in **Ansys SIwave**, ensuring precise **50-ohm trace impedance** matching across the board.

Modeling and Signal Integrity Analysis of PCB Topologies | *Ansys 2D Extractor, Nexxim*

- Modelled **50-ohm** and **100-ohm impedance matched** *microstrip, stripline, and differential pair topologies*
- Verified **EMI containment** by simulating and analyzing 2D electric and magnetic fields across all channel cross-sections.
- Evaluated **1 Gbps signal integrity** over a 100 mm trace using transient circuit model to perform dynamic **eye diagram analysis.**

High-Speed PCB Signal Integrity Across Multiple Topologies | *Altium Designer* | *Ansys SIwave* | *PCB Design*

- Designed **3 PCB topologies** (Coplanar, Stripline, Differential) in Altium Designer, tuning trace widths for **50-ohm** impedance.
- Extracted **S-parameter** matrices using **Ansys SIwave** to establish baseline limits for **-3dB insertion loss** and **-10dB return loss**.
- Verified board performance via **VNA**, mapping experimental -3dB insertion loss cutoff **(3.19 GHz)** against simulated **(4.29 GHz).**

Computational Analysis of Microelectronic Heat Sinks | *ANSYS Icepak*

- Simulated **3D** steady-state CFD models of **36-fin** aluminum **heat sinks** cooling an **8 W** silicon **microchip** in a confined cabinet.
- Achieved a **<1% thermal variance** accuracy by conducting a **3-stage mesh refinement** study up to 678k hexahedral elements
- Evaluated the **engineering trade-off** between thermal dissipation and aerodynamic resistance across **4 heat sink** configurations.